

Technical University of Munich
Chair of AI Planning in Dynamic Environments

CIT433020– Final project
Contact-Rich Robotics; an Optimization Perspective

Instructor: Majid Khadiv

Due date: 22.07.2025 (1pm)

This project has 4 phases. The first phase is a bonus, so you could skip it if you do not want the bonus :) The total of points for the other phases is 50 (with 50 from the midterm results in your final grade of 100). You need to write a report and submit it together with the code by 22nd of July (before 1pm). Furthermore, you need to prepare a 10-minute presentation about your results and discuss them in the class on the 22nd or 23rd of July. Good luck!

If you have any questions regarding the theory, you can contact majid.khadiv@tum.de. For questions regarding Pinocchio and Casadi, contact michal.ciebielski@tum.de. For questions about MuJoCo, contact victor.dhedin@tum.de. For questions regarding RL, contact shafeef.omar@tum.de, and finally for questions related to sampling-based MPC contact ilyass.taouil@tum.de.

Distribution of Marks

Question	Points	Score
1	20	
2	20	
3	20	
4	10	
Total:	70	

In the final project, the goal is to control a finger robot shown in Fig. 1 to push the red box to the final (green) target (use the robot model from the MuJoCo tutorial session). As discussed in the class, to control the robot, we need to build a forward simulation of the robot with contact.

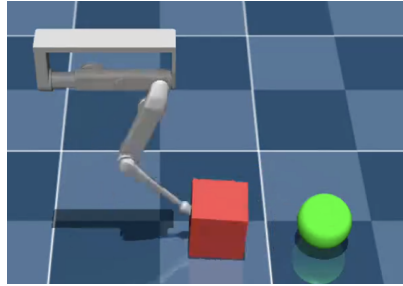


Figure 1: Finger pushing a box (red) to a final target (green)

1. In the first phase of the project, we develop our own simulation environment with rigid contact, using different solvers. This example can help you set up a robot, find unconstrained dynamics, and also detect collisions.
 - (a) (8 points) Implement the Projected Gauss-Seidel (PGS) algorithm to solve the contact complementarity problem formulated as LCP, NCP, and CCP. Compare the behavior of the system with each of the methods. For the tests, use a simple trajectory of the robot that hits the box.
 - (b) (7 points) Now solve the CCP problem using the relaxation proposed by MuJoCo (add regularization to the Delassus matrix). Use Casadi to solve the optimization problem. Compare the results with the PGS solver for CCP.
 - (c) (5 points) Compare the results of your simulation to the MuJoCo simulation and discuss your most interesting findings.
2. Now we would like to find a control policy using RL that pushes the box to a desired final location. For this, we will use the MuJoCo simulation environment with the Gymnasium interface.
 - (a) (10 points) Find a policy that pushes the object to a desired target (green in Fig. 1) in the forward direction.
 - (b) (5 points) Now find a policy that pushes the object to the desired location, without rolling the object (the orientation of the object remains the same).
 - (c) (5 points) Analyze the results and comment on your findings.
3. Now we would like to apply sampling-based MPC that pushes the box to a desired final location. For this example, we use the MuJoCo simulation environment.
 - (a) (10 points) Find a policy that pushes the object to a desired target (green in Fig. 1) in the forward direction.
 - (b) (5 points) Now find a policy that pushes the object to the desired location, without rolling the object (the orientation of the object remains the same).

- (c) (5 points) Analyze the results and comment on your findings.
- 4. In the last part of the project, the goal is to compare the results from the MPC and RL controllers
 - (a) (10 points) compare the MPC and RL controllers and describe which one would be your choice in the future. Explain your answer.